

An Open Source Number Theoretic Transform Accelerator for ML-KEM-512

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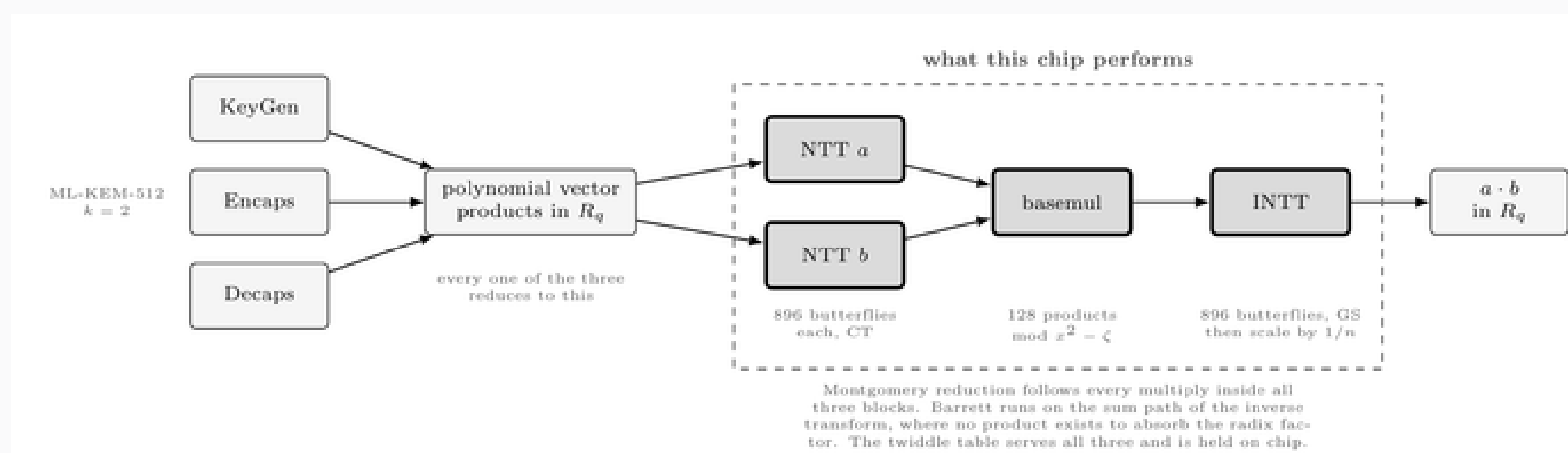
Motivation

- ML-KEM (FIPS 203) is the NIST standard for post quantum key encapsulation. Its cost is dominated by polynomial multiplication in $\mathbb{Z}_{3329}[x] / (x^{256} + 1)$, and the Number Theoretic Transform is what makes that multiplication fast enough to use
- Almost every published NTT accelerator is designed and verified on proprietary EDA tools whose licences are costly, thus limiting access to students who would want to design their own ASICs

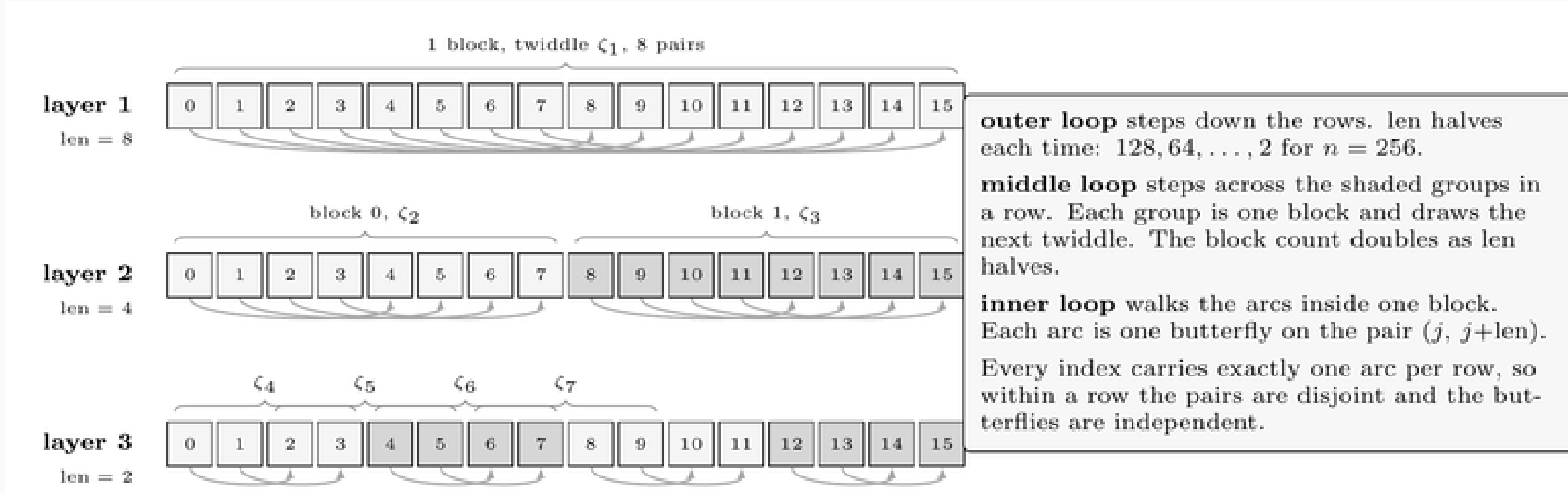
Our Goal for this Project:

Build a complete ML-KEM-512 forward and inverse transform on an open PDK, SKY130, that runs a full RTL-GDSII ASIC flow for students, hobbyists, and educators to learn from and edit for their liking

The Algorithm

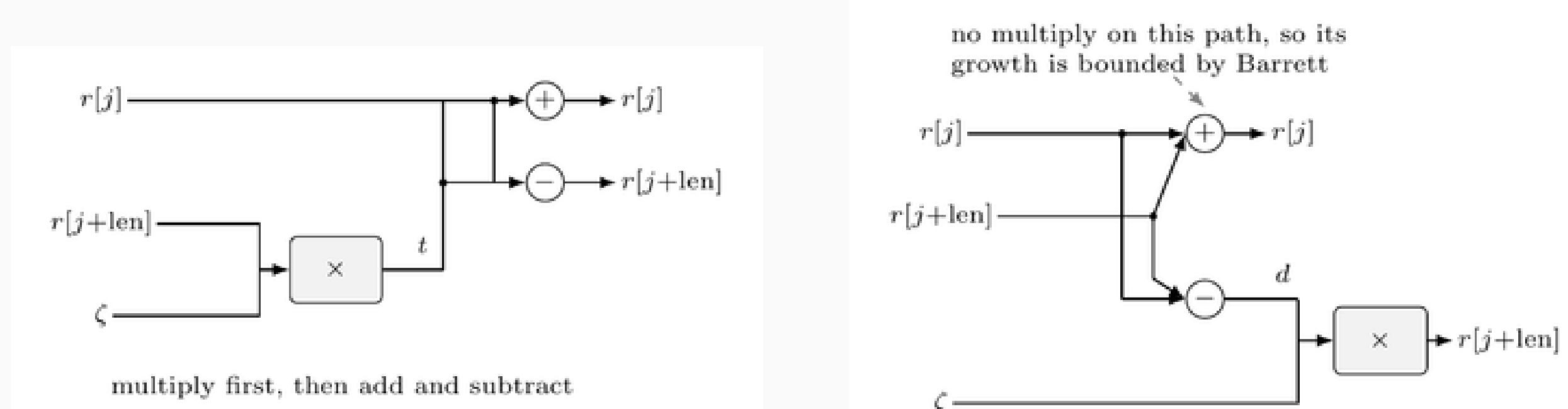


Every ML-KEM-512 polynomial product runs as a forward transform on each operand, 128 small products in the transform domain, then one inverse transform.



The transform is three nested loops. The outer loop steps through the 7 layers, halving the stride each time. The middle loop walks the blocks in a layer and draws one new twiddle factor per block. The inner loop issues the butterflies, which are independent inside a layer.

All 128 twiddle factors sit in a table on the die, so the host sends an index and never a constant.

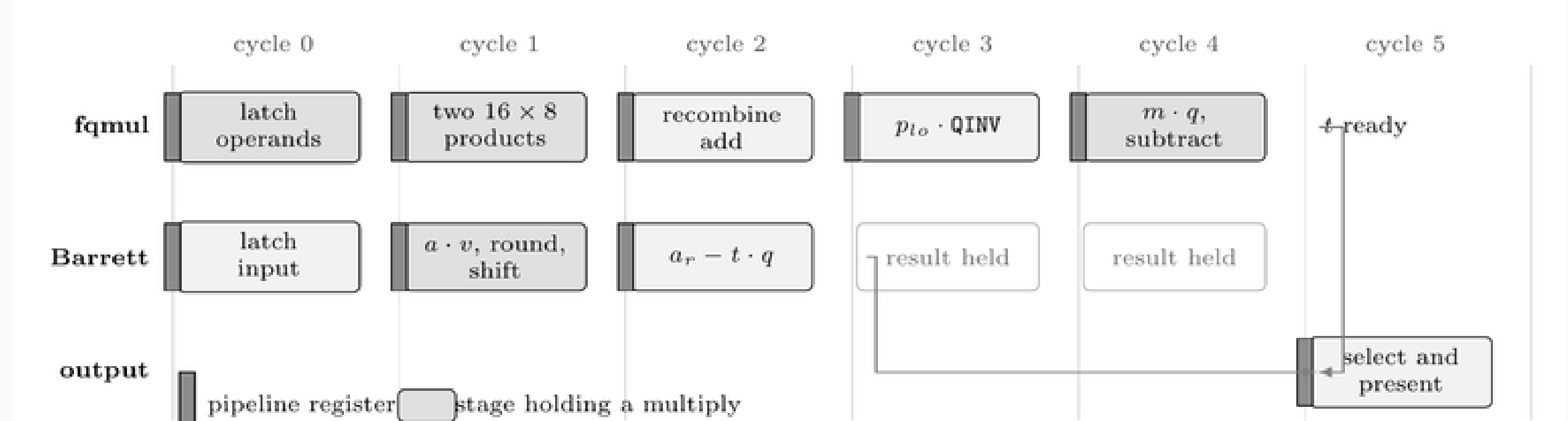


Cooley Tukey butterfly (forward). Multiply by ζ , reduce with Montgomery, then add and subtract.

Gentleman Sande butterfly (inverse). Add and subtract first, reduce the sum with Barrett, then multiply the difference by ζ and reduce with Montgomery.

The Datapath

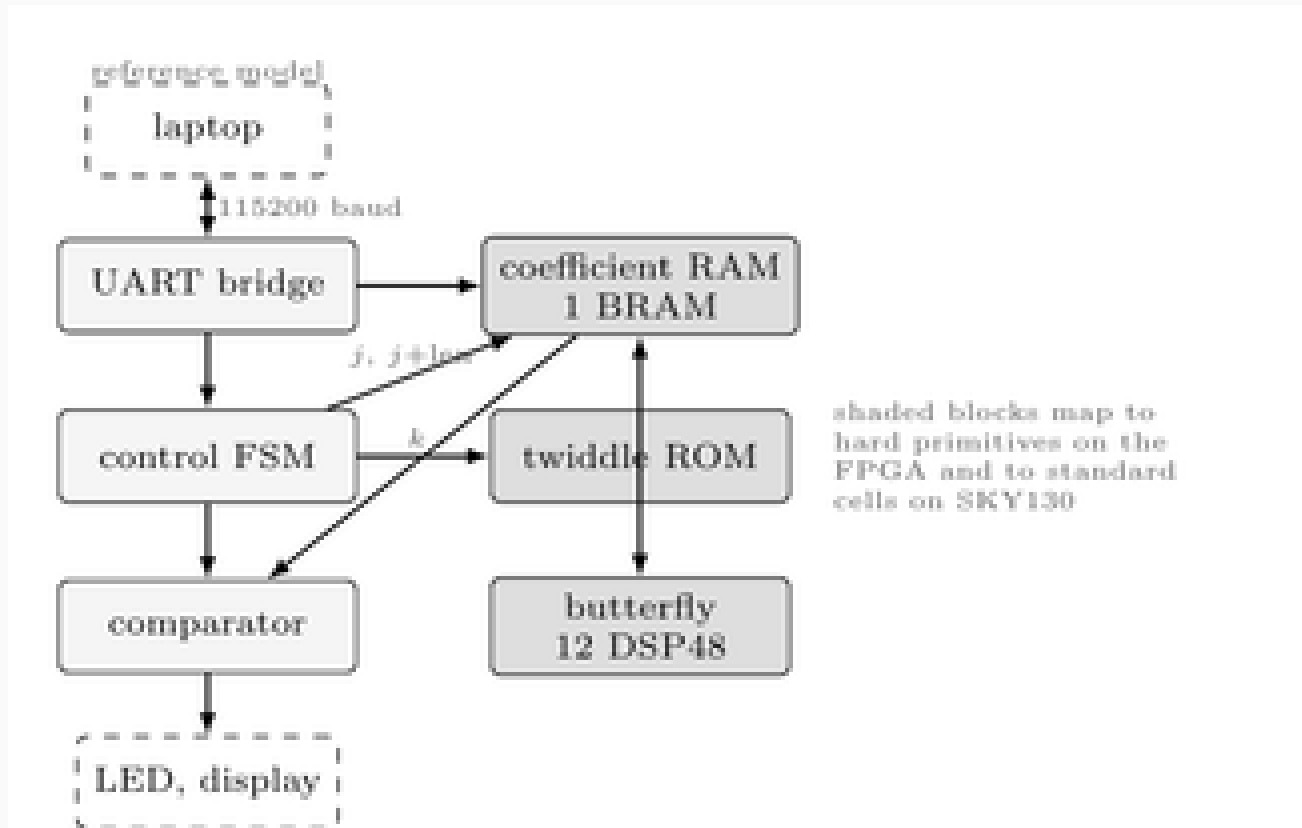
The chip is the modular arithmetic engine of the transform. A host holds the 256 coefficients and walks the address pattern, the chip does the arithmetic and looks up its own constants. In total, six operations cover the whole flow, with every one of them taking 5 clocks.



Both reduction kernels are on the die. Montgomery follows every multiply, Barrett runs on the addition path of the inverse transform, where there is no product to absorb the radix factor. The multiplier is pipelined across five stages, with two 16 by 8 partial products in place of one 16 by 16, which is what closed timing without a hard multiplier block.

FPGA Demonstration

FPGA Demo Diagram

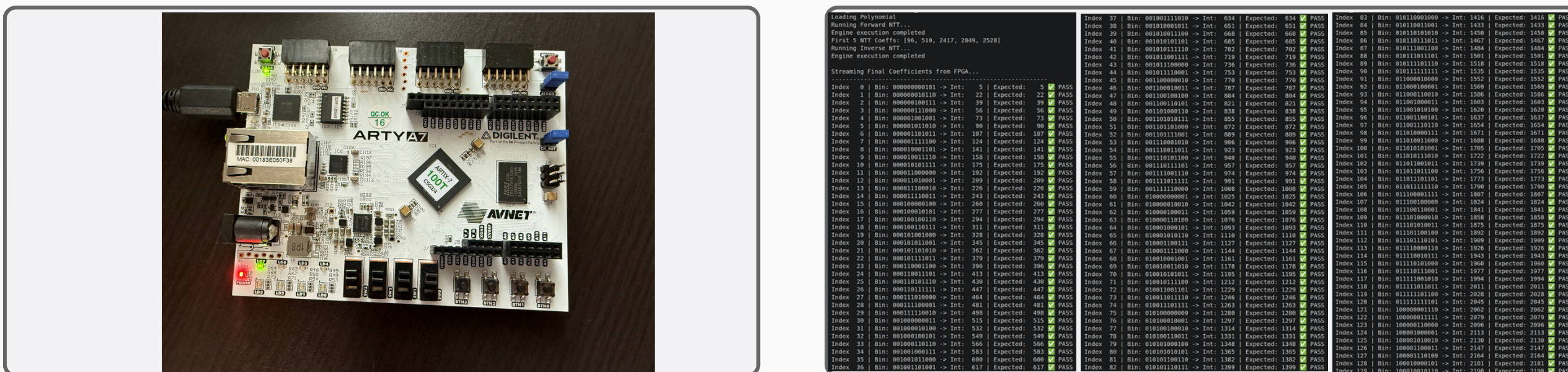


Measured on hardware, post route

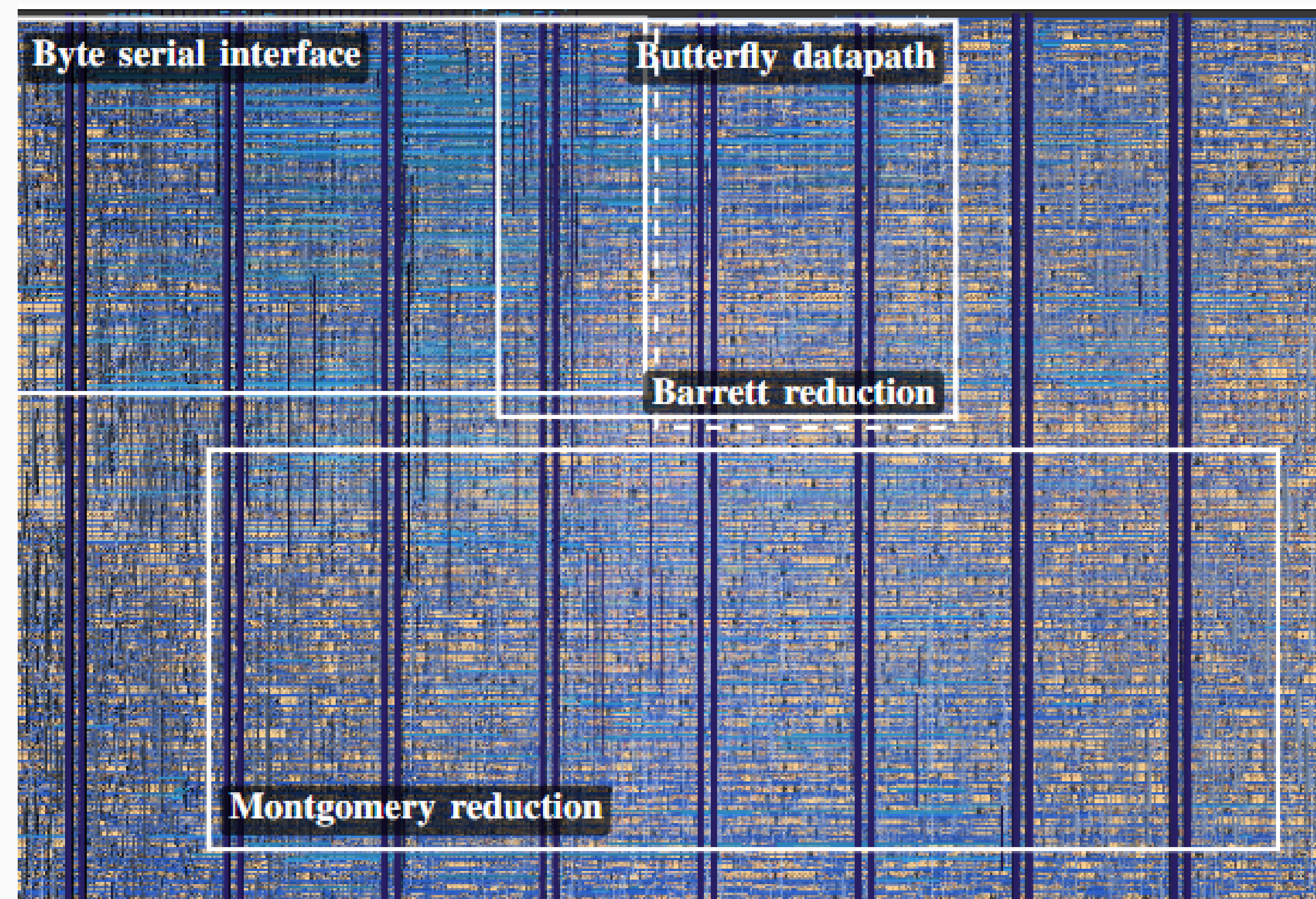
Arty A7-100T	Used
Slice LUTs	532 (0.84 %)
Slice registers	284 (0.21 %)
DSP48E1	9 (3.75 %)
Block RAM (RAMB18)	1 (0.37 %)
Bonded IOB	8 (3.81 %)
Global buffer	1 (3.13 %)
Device	xc7a100tcsq324-1

Post route result	Value
Clock	100 MHz
Worst setup slack	+0.960 ns
Worst hold slack	+0.176 ns
Failing endpoints	0 of 922
Implied maximum	110.6 MHz
Total on chip power	0.119 W
Dynamic / static	0.022 / 0.097 W

Live demonstration



The Silicon

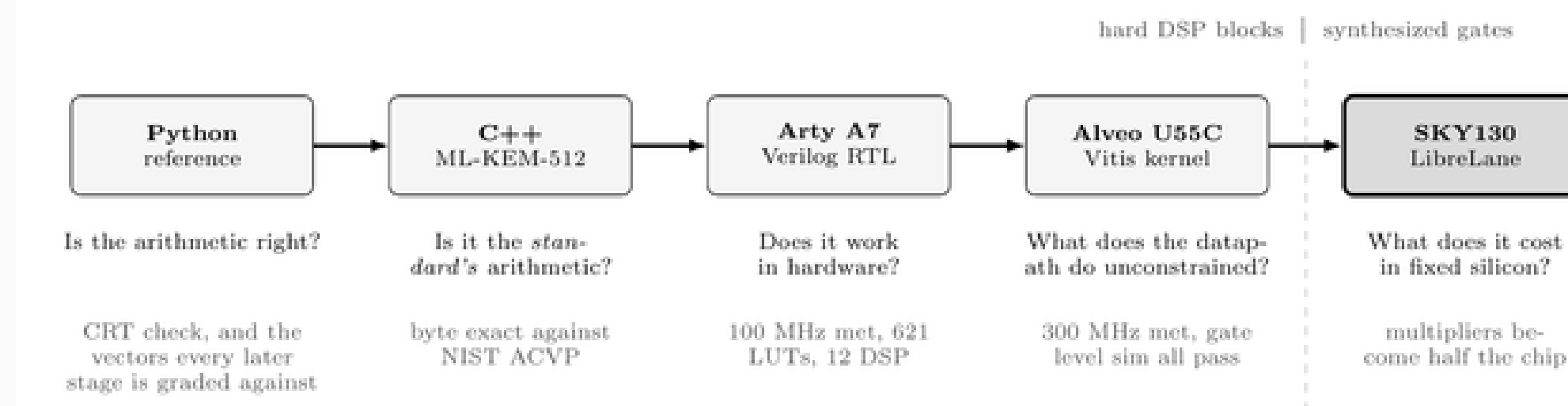


We hardened the design with LibreLane on SkyWater SKY130, inside a 2x2 Tiny Tapeout tile. All 6,067 cells are standard cell logic, and the arithmetic is 72 percent of them. Each flip flop is attributed to the module that owns the register driving it, thus this is how the blocks were located

Technology	SKY130, 130 nm	Target clock	50 MHz
Cell library	sky130_fd_sc_hd	Max frequency	55.4 MHz
Supply	1.80 V	Setup slack	+1.955 ns
Die area	334.88 x 225.76 μm^2	Hold slack	+0.106 ns
Logic area	52,731 μm^2	Violations	0 at 9 corners
Cell count	6,067	DRC / LVS	0 / 0
Gate count	14,048 GE	Total power	2.623 mW
Utilization	75.5 %	Routed wire	128,063 μm

The design closes 50 MHz with 1.955 ns of setup slack and zero violations at all nine corners, which implies 55.4 MHz at the slow corner. DRC, LVS, antenna and inferred latch counts are all zero.

From Model to Silicon



Each stage is checked against the one before it, so a disagreement always has a known reference. The dashed line marks the boundary that mattered, for the multipliers are hard DSP blocks on both FPGAs and synthesized gates on SKY130, thus they become half the chip.

Future Works

- Given more time, we would have liked to improve our processing power by using a cpu on chip that would be able to store and process the twiddle ROM coefficients
- The table already sits in silicon, but nothing on the die decides what to do with it, so the host holds all 256 coefficients and walks the FIPS 203 address pattern itself